ASRC Fuel Tank Robot Daily Report

Model data, prototype files and the round-trip cycle with 3 N pretension, 11 November 2026

Today started from the team's new length-controlled round-trip study and the design limitation list on main. In that study, no scenario passed every check: the nominal cycle had an unstable startup and rope tension dipping to 0.71 N. Today's work kept the model parameters unchanged and fixed the rope-length commands.

Summary of today's work

- **Reviewed the newest results.** Round-trip cycle (straighten, insert, sweep right and back, sweep left and back, withdraw) and the design limitations. The failures came from the commands (pretension margin, startup) and from open-loop calibration, not from the geometry.
- **Model data sheet.** All model inputs exported from the code: continuous 96 mm modules with no pin joint, one 1.8 mm NiTi rod along the full length of each M module, module mass, EA, EI and torsion, rope, winch, tank and numerical settings (model_data/MODEL_DATA.md, JSON and CSV).
- **Prototype files marked.** The moving_prototype/full_length files (84 mm shells, 12 mm pin joints, separate rods with a free joint) do not match the design and are marked to be ignored. Nothing in them was changed apart from the notice.
- **Stable startup.** The head is raised from natural sag (minus 57 degrees, all ropes at 3 N) to 20 degrees through held states with every rope at 3 N or more. All 32 states are stable (peak rope 17.4 N).
- **Round trip with 3 N pretension.** Same parameters, new commands. The nominal cycle now passes every check: 0 of 1069 samples fail (11 November: 56 of 1033). The rope EA minus 20% case also passes. The other three sensitivity cases fail only on open-loop calibration.

Case (same open-loop commands)	11 Nov: 1 N, payout start	11 Nov: 3 N, stable startup	Remaining failure type
Nominal	56 of 1033 fail	**0 of 1069 fail**	none
Rope EA -20%	1020 of 1020 fail	**0 of 1071 fail**	none
Free rope 1.5 m	1023 of 1023 fail	**543 of 1070 fail** (clean to 0 deg)	rope below 0.95 N (543)
Rod E 75 GPa	1019 of 1019 fail	**318 of 1063 fail** (clean to 32 deg)	yaw error > 1 deg (278), elevation error > 5 deg (180)
Head 550 g	1009 of 1009 fail	**262 of 1067 fail** (clean to 78 deg)	yaw error > 1 deg (230), elevation error > 5 deg (208)

Decisions from today

Use a 3 N minimum on every rope and the held-state startup. Do not raise the floor to 5 N without finer pose spacing. Treat the remaining failures as a calibration and feedback problem: add head-angle and rope-tension feedback, or recompute commands from measured stiffness, head mass and rope data. Ignore the full_length prototype files.

Round trip with 3 N pretension and a stable startup

Model parameters are unchanged from the data sheet: U4+M2, continuous 96 mm modules, full-length 1.8 mm NiTi rod at E 40 GPa, 500 g head, PA6 ropes with 1 m free length, 20 degree installation, anchors [4,5,6], 8 cells per module. The cycle script and acceptance checks are the team's from this morning; only the rope-length commands are new.

- **Startup (startup_sweep.py).** Start outside the tank (clamp s = minus 750 mm) in natural sag with every rope at 3 N. Raise the held head elevation in 2.5 degree steps to 20 degrees, minimising the largest rope force with all ropes at 3 N or more; every state is checked for stability, geometry, strain and elevation on the 8-cell mesh.
- **Yaw sweep (min_force_sweep.py).** Minimum-force poses every 5 degrees from 0 to 175 degrees with the head held at 20 degrees and every rope at 3 N or more, starting from the startup end state.
- **Cycle (length_cycle.py).** Rope lengths of these states drive all 12 winches; each step is solved from the previous state. Acceptance: rope 0.95 to 250 N, stable, tank clearance at least 5 mm after a 1.5 mm allowance, yaw error at most 1 degree, elevation 20 plus or minus 5 degrees.

Nominal cycle, 3 N	Result
Samples / failures	1069 / 0
Yaw reached with all checks passing	0 to 175 deg, right and left
Minimum / peak rope tension	2.74 N / 17.6 N
Minimum stability eigenvalue	0.166 (11 Nov: -0.230, unstable)
Minimum tank clearance	20.5 mm (19.0 mm after allowance)
Largest yaw / elevation error	0.24 deg / 3.93 deg
Head return error at zero yaw	0.0000 mm

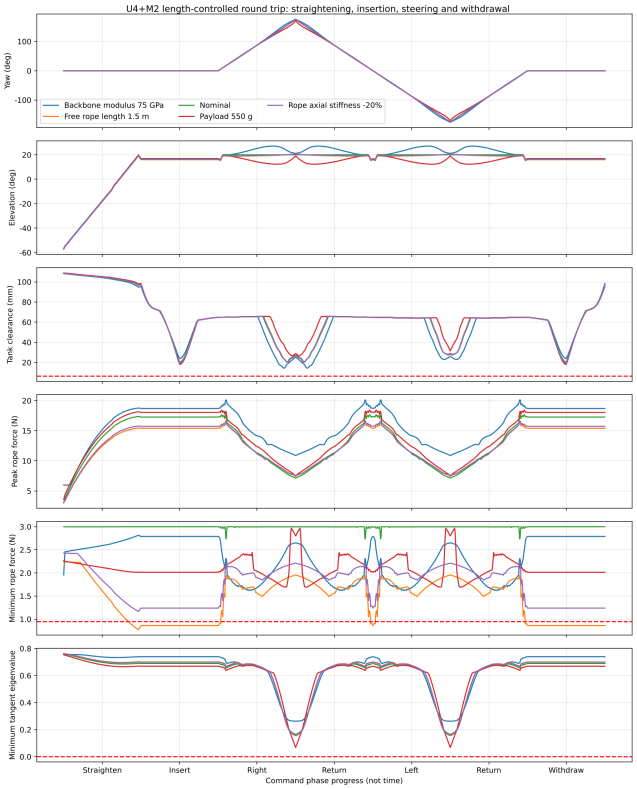


Figure 1 Round trip with 3 N pretension, all five cases: yaw, elevation, tank clearance, peak and minimum rope force, stability. Red dashed lines are acceptance limits.

Sensitivity cases and the 5 N comparison

Case	3 N: failures	Min rope N	Max yaw / elev error, deg	Clean outbound yaw	5 N: failures
Nominal	0	2.74	0.2 / 3.9	175 deg	48
Rope EA -20%	0	1.17	0.7 / 4.6	175 deg	48
Free rope 1.5 m	543	0.77	0.9 / 4.8	0 deg	52
Rod E 75 GPa	318	1.62	6.3 / 7.2	32 deg	408
Head 550 g	262	1.69	9.0 / 7.7	78 deg	310

- **No case has an unstable state, and the startup is stable in all five.** The startup and stability problems of the original cycle are removed.
- **Remaining failures are open-loop calibration.** A stiffer rod or heavier head moves the head off target (up to 6 to 9 degrees). A longer free rope stretches less for the same commands, so tension drops below 0.95 N, mainly during insertion and withdrawal. More pretension does not fix these; feedback or recalibrated commands do.
- **5 N is worse than 3 N.** The 5 N pose sequence re-allocates 13.4 N between the 50 and 55 degree poses, and length interpolation cannot follow it: all 48 nominal failures lie between 51.9 and 53.9 degrees. The 3 N sequence changes at most 5.8 N between poses. 3 N is the recommended floor.
- **Known imperfection.** During insertion and withdrawal the head sits at about 17 degrees, not 20, because the 0 degree pose of the 3 N sweep was accepted on the fine mesh within the plus or minus 5 degree band. It is inside the acceptance limit; requiring exact elevation on the fine mesh would remove it.

Limitations

- Quasi-static, frictionless model: no dynamics, rope friction, winch backlash, NiTi hysteresis or fatigue. Rod properties are calculated, not measured.
- The head is a 500 g point mass with a 63 mm envelope; actual head CAD and external cables are not included. Withdrawal is rigid translation of the front.
- Sensitivity cases replay identical commands for assumed parameter changes; they are not measured probabilities.

Next steps

Priority	Task	Purpose
1	Closed-loop cycle simulation	Correct rope lengths from head angle and rope tension (sensitivity matrix from 10 Nov); re-run the 75 GPa, 550 g and 1.5 m cases to within 1 deg and 0.95 N
2	Exact elevation at every pose	Tighten the 8-cell acceptance of the yaw-sweep poses so insertion holds 20 deg
3	Build and test the backbone M module	Continuous 96 mm module with full-length rod: EA, EI, buckling, end-joint slip; replace assumed values
4	Rope and winch calibration	Measured rope EA, free length and friction; command recalibration
5	Head CAD, entry, storage, rear mechanism	As in the design limitation list

Scripts and data: ASRC_kevin/telescopic_planar_study/backbone_20261108 (README section 10; startup_sweep.py, min_force_sweep.py --seed-startup, length_cycle.py --source/--startup/--out, summarize_cycle_run.py; results in length_cycle_results_pre3 and length_cycle_results_pre5; model_data/MODEL_DATA.md). This report is generated by ASRC_kevin/telescopic_planar_study/reports/build_report_2026_11_11.py.